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SIMATS ENGINEERING

ASSESSMENT TOOL 5

COURSE NAME:Artificial Intelligence

COURSE CODE: CSA1715

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 5

Total Marks:50

Annexure A

CONSTRAINT SATISFACTION PROBLEM

Code of Conduct:

I _K.YAMINI(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

CONSTRAINT SATISFACTION PROBLEM:-

In constraint satisfaction problem , the goal state of the problem is achieved by satisfying a given

set of rules.

■ A constraint satisfaction problem (or CSP) is defined by:

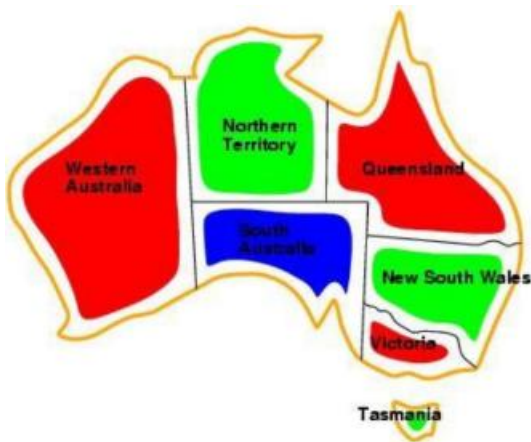
- a set of variables, $X_1, X_2, \dots, X_n$, and
- a set of constraints, $C_1, C_2, \dots, C_m$.
- Each variable X_i has a nonempty domain D_i of possible values.
- A complete assignment is one in which every variable is mentioned, and a solution to a CSP is a complete assignment that satisfies all the constraints.

Example: Map-Coloring:



- Variables WA, NT, Q, NSW, V, SA, T
- Domains $D_i = \{\text{red, green, blue}\}$
- Constraints: adjacent regions must have different colors
- e.g., $WA \neq NT$,
- or $(WA, NT) \in \{(\text{red, green}), (\text{red, blue}), (\text{green, red}), (\text{green, blue}), (\text{blue, red}), (\text{blue, green})\}$

Example: Map-Coloring



- Solutions are complete and consistent assignments, e.g.,

■ WA = red, NT = green, Q = red, NSW = green, V = red, SA = blue, T = green

Examples of this sort of problem include cryptarithmic puzzles many real-world perceptual labeling problems.

Design tasks can also be viewed as constraint-satisfaction problems in which a design must be

created within fixed limits on time, cost and materials.

Constraint satisfaction is a two-step process. First, constraints are discovered and propagated as far as possible throughout the system. Then, if there is still not a solution, search begins. A guess about something is made and added as a new constraint. Propagation can then occur with

this new constraint, and so forth.

The first step, propagation, arises from the fact that there are usually dependencies among the constraints.

These dependencies occur because many constraints involve more than one object and many objects participate in more than one constraint. So, for example, assume we start with one constraint, $N = E + 1$.

Then, if we added the constraint $N = 3$, we could propagate that to get a stronger constraint on E. namely that $E = 2$.

Constraint propagation also arises from the presence of inference rules that allow additional constraints to be inferred from the ones that are given. Constraint propagation terminates for one of two reasons.

First, a contradiction may be detected. If this happens, then there is no solution consistent with all the known constraints. If the contradiction involves only those constraints that were given as part of the problem specification {as opposed to ones that were guessed during problem

solving), then no solution exists.

Once this has been done, constraint propagation can begin again from this new state. If a solution

is found, it can be reported. If still more guesses are required, they can be made. If a contradiction

is detected, then backtracking can be used to try a different guess and proceed with it.

Algorithm:Constraint satisfaction

1. Propagate available constraints. To do this, first set OPEN to the set of all objects that must have values assigned to them in a complete solution. Then do until an inconsistency is detected or until OPEN is empty:
 - (a) Select an object OB from OPEN. Strengthen as much as possible the set of constraints that apply to OB.
 - (b) If this set is different from the set that was assigned the last time OB was examined or if this is the first time OB has been examined, then add to OPEN all objects that share any constraints with OB.
 - (c) Remove OB from OPEN.
 2. If the union of the constraints discovered above defines a solution, then quit and report the solution.

If the union of the constraints discovered above defines a contradiction, then return failure.
 3. If neither of the above occurs, then it is necessary to make a guess at something in order to proceed. To do this, loop until a solution is found or all possible solutions have been eliminated:
 - (a) Select an object whose value is not yet determined and select a way of strengthening the constraints on that object.
 - (b) Recursively invoke constraint satisfaction with the current set of constraints augmented by the strengthening constraint just selected.
- is a problem state in which all letters have been assigned a digit in such a way that aJI the initial constraints are satisfied.

Problem:

SEND
+ MORE

.....
MONEY

Initial State:

No two letters have the same value.
The sums of the digits must be as shown in
the problem.

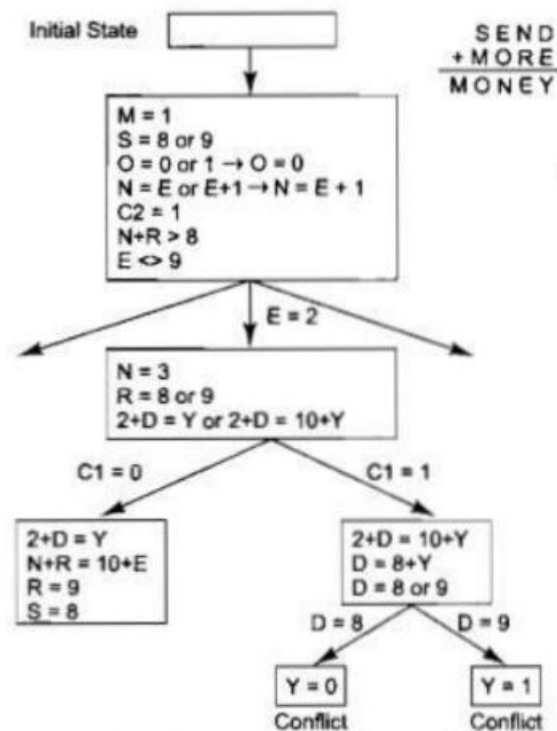
A Cryptarithmic Problem

The solution process proceeds in cycles. At each cycle, two significant things are done (corresponding to steps I and 4 of this algorithm):

1. Constraints are propagated by using rules that correspond to the properties of arithmetic.
2. A value is guessed for some letter whose value is not yet determined.

Initially, rules for propagating constraints generate the following additional constraints:

- $M = 1$, since two single-digit numbers plus a carry cannot total more than 19.
- $S = 8$ or 9 , since $S + M + C_3 > 9$ (to generate the carry) and $M = 1$, $S + 1 + C_3 > 9$, so $S + C_3 > 8$ and C_3 is at most 1.
- $O = 0$, since $S + M(1) + C_3 (\leq 1)$ must be at least 10 to generate a carry and it can be at most 19. But M is already 1, so O must be 0.
- $N = E$ or $E + 1$, depending on the value of C_2 . But N cannot have the same value as E . So $N = E + 1$ and C_2 is 1.



Solving the cryptarithmic puzzle

- In order for $C2$ to be 1, the sum of $N + R + C1$ must be greater than 9, so $N + R$ must be greater than 8.

- $N + R$ cannot be greater than 18, even with a carry in, so E cannot be 9.

At this point, let us assume that no more constraints can be generated. Then, to make progress from here, we must guess. Suppose E is assigned the value 2. (We chose to guess a value for E because it occurs three times and thus interacts highly with the other letters.) Now the next cycle begins.

The constraint propagator now observes that:

- $N = 3$, since $N = E + 1$. Again, assuming no further constraints can be generated, a guess is required. Suppose $C1$ is chosen to guess a value for. If we try the value 1, then we eventually reach dead ends, as shown in the Fig .. When this happens, the process will backtrack and try $C1 = 0$.

A second thing to notice is that there are often two kinds of constraints.

The first kind is simple; they just list possible values for a single object.

The second kind is more complex; they describe relationships between or among objects. Both kinds of constraints play the same role in the constraint satisfaction process, and in the cryptarithmic example they were treated identically.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total (50)
Understanding of Scenario	20% (2)	/2	/2	/2	/2	/2	/10
Identification of Key Issues	20% (2)	/2	/2	/2	/2	/2	/10
Application of Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/2.5	/12.5
Decision Making	20% (2)	/2	/2	/2	/2	/2	/10
Clarity of Response	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/1.5	/7.5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/50

1. Intelligent Agents

1. Understanding of Scenario:

An intelligent agent is a system that perceives its environment using sensors and takes actions using actuators to achieve specific goals. It acts rationally based on given conditions.

2. Identification of Key Issues:

- How the agent collects information (sensors)
- How it processes the data
- How it selects the best action
- Measuring performance (efficiency, accuracy)

3. Application of Concepts:

Types of intelligent agents:

- Simple Reflex Agent: Acts based on current perception
- Model-Based Agent: Maintains internal state
- Goal-Based Agent: Works to achieve a goal

- Utility-Based Agent: Chooses best action based on utility

Example: Vacuum cleaner agent cleans room when dirt is detected.

4. Decision Making:

The agent uses condition-action rules or algorithms to decide actions. For example, if dirt is present → clean; else → move.

5. Clarity of Response:

Thus, intelligent agents are the core of AI systems and are used in robotics, automation, and real-world applications.

2. Uninformed Search Algorithms

1. Understanding of Scenario:

Uninformed search algorithms solve problems without any prior knowledge of the goal. They explore the search space blindly.

2. Identification of Key Issues:

- High time complexity
- Large memory usage
- No guidance toward goal
- May explore unnecessary paths

3. Application of Concepts:

Main algorithms:

- Breadth First Search (BFS): Explores level by level
- Depth First Search (DFS): Explores deeply first
- Uniform Cost Search (UCS): Chooses lowest cost path

4. Decision Making:

- Use BFS when shortest path is needed
- Use DFS when memory is limited

- Use UCS when costs differ

5. Clarity of Response:

Uninformed search is simple but inefficient compared to informed search methods.

3. Water Jug Problem

1. Understanding of Scenario:

The problem is to measure exactly 2 liters using a 4-liter jug and a 3-liter jug.

2. Identification of Key Issues:

- Representation of states (x, y)
- Valid operations (fill, empty, transfer)
- Achieving goal state

3. Application of Concepts:

Using state space search:

Steps:

$(0,0) \rightarrow \text{Fill 4L} \rightarrow (4,0)$

$(4,0) \rightarrow \text{Pour to 3L} \rightarrow (1,3)$

$(1,3) \rightarrow \text{Empty 3L} \rightarrow (1,0)$

$(1,0) \rightarrow \text{Transfer} \rightarrow (0,1)$

$(0,1) \rightarrow \text{Fill 4L} \rightarrow (4,1)$

$(4,1) \rightarrow \text{Pour} \rightarrow (2,3)$

Goal achieved: 2 liters

4. Decision Making:

Breadth First Search is used to find shortest path solution.

5. Clarity of Response:

Thus, the problem is solved efficiently using AI search techniques.

4. 8-Queens Problem

1. Understanding of Scenario:

Place 8 queens on a chessboard such that no queen attacks another.

2. Identification of Key Issues:

- No two queens in same row
- No two queens in same column
- No two queens in same diagonal

3. Application of Concepts:

Use Backtracking Algorithm:

- Place queen in first row
- Check if safe
- Move to next row
- If conflict occurs → backtrack

4. Decision Making:

Backtracking avoids unnecessary computations by rejecting invalid solutions early.

5. Clarity of Response:

Thus, the 8-Queens problem is a classic example of constraint satisfaction problem.

5. AI Application – OLA Cab System

1. Understanding of Scenario:

OLA uses AI to match passengers with nearby drivers.

2. Identification of Key Issues:

- Finding nearest driver
- Estimating time and distance
- Handling traffic conditions

- Efficient allocation

3. Application of Concepts:

- GPS tracking
- Search algorithms
- Machine learning for prediction
- Route optimization

4. Decision Making:

System selects driver based on shortest distance and minimum time.

5. Clarity of Response:

Thus, AI improves efficiency, reduces waiting time, and enhances user experience.